Directions for *Drosophila* Longevity Experiment 🍌 🪰

Crosses:

(Customize for your genotypes and experimental conditions)

- 12-15 female Arm>Gal4 virgins with 2-5 males from stocks of interest. Cross in triplicates.
    - 3 controls: two negative controls of Arm>Gal4 x VDRC 60000, Arm>Gal4 x VDRC 2691. A positive control like Arm>Gal4 x VDRC 11371.
- Leave at 25℃ for 3 days
- Transfer to lightly yeasted bottles at 25℃
- Remove parents approximately 7 days after setting up the crosses. Monitor egg laying and allow additional time if necessary to ensure adequate progeny production.
- Collect at around ~1 week to 12 days after transferring.
    - Monitor for eclosion. Kill early emergers day before scheduled collection

Collection:

Highly suggest using male and female flies – sex is an important biological variable for lifespan. Notably, virgin and non-virgin females have observed differences in lifespan. Plan your experiment accordingly.

- Collect males AND females separately into vials
    - Preferably collect female virgins
- Approximately 60-120 males and 60-120 females, 10 per vial
- Set timer for ~9 minutes for collection at the microscope to limit amount of time each bottle is exposed to CO2. Flies are sensitive to CO2 exposure, and this can shorten lifespan.

Keep vials organized in a tray. Make sure tray is exposed to light conditions for your experiment.

Monitoring:

Change food MONDAY/WEDNESDAY/FRIDAY, or AT LEAST 3 times a week. Use Lifespan Experiment CSV template sheet to record data while changing trays.

NOTE: Limit the amount of time flies are kept outside of the 25℃ incubator! Do not leave extra trays idling outside incubator while changing out food.

Recording Data – Template Format:

These experiments were conducted in 96 vials stored in an 8x12 tray format. Therefore, the given template is formatted for this experimental set-up. Template can be easily customized to account for your experiment's design.

Genotype: Records the genotype contained in each vial. I recommend recording this data separately and leaving the column blank until final data analysis. This ensures the experiment is blinded.

ID: This is a unique ID assigned to each fly within a given vial. Typically, each vial has 10 flies, and so 10 IDs exist per vial (Example: A1-1 to A1-10).

Vial: Name of the container flies are kept in during the experiment. This can be whatever name you would like. I typically use vial identifier, based on position in the tray. In the template, vials are named A1-A12, B1-B12, and so on through H1-H12.. Please note that Excel may order these non-sequentially, so A1 is followed by A10, A11, A12, then A2 through A9. You may need to use the search function to find a vial, or re-format the template to your needs.

Day: Day of Death column. In this column, record a day matching the Tray Day for a fly ID when a fly dies.

Escape: This is used to track the escape of any flies.

Directions:

1) Find the template sheet tab that matches the tray number
2) Change the Day on the tray, make sure to double check the day is accurate to Day 1.
3) Transfer flies CAREFULLY into new tubes. Try to avoid fly escapes (see #8 for directions if any flies get loose)
4) Check bottom of the old tube for flies stuck in the tube. Carefully observe the legs or wings for movement showing fly is alive. If a stuck fly is alive, use a paint brush to gently dislodge and transfer to the new tube, ensuring fly reaches the bottom of the new tube and isn't stuck on the walls.
    a. Use a second new vial if needed. Transfer flies into a new vial, place stuck fly carefully in the second vial, then transfer the other flies into that same second vial. It's very important to avoid letting flies escape to the best of your ability.
5) If the fly at the bottom of the old tube is dead, record it as dead in the corresponding Vial position of the excel sheet. Scroll to the tube ID (Example: A7 under Vial) and find the next empty slot under "Day" that has no data recorded (Example: A7-1 might have day 2 recorded, A7-2 might have Day 20 recording, but A7-3 is blank and is the next ID to record data for). Then, record the Day of the tray in the Day of Death column. In the Escape column, enter "1" to indicate that the fly died naturally and should be included as an observed death (i.e., not censored).
6) DO NOT record a fly in step 5 if a fly was labeled as "Previous Day carried" (Example: 30 carried). This means the fly was previously recorded as dead and was accidentally transferred to the new tube. Instead, delete "carried" from the previous date's entry.

7) Check the bottom of the NEW tube that flies were transferred into. Ensure all flies in tube are living. Record any dead flies as described above in #5, but record these as “Day carried” (Example: 28 carried)
8) If any flies escape, instead of recording an entry as dead, put “0” in the “Escape” column for the number of flies that flew out of the vial. This will censor the fly from the experiment. You can record either the day or “0” in the Day column.
9) If any flies are accidentally crushed during transfer, enter “0” in the Escape column to indicate that the fly should be censored rather than counted as a natural death.

Once experiment is complete, the R script can be used to analyze results.

Analysis:

1) Ensure CSV dataset contains no empty entries. Properly censor any flies that were lost or otherwise removed from the experiment*.
2) Use flyfespan.R to analyze complete CSV dataset
3) Run all preliminaries under “Run this BLOCK first”. Ensure all dependencies load properly.
4) Edit custom data including your p-value cutoff, control, and longevity assay CSV file (suggested P<0.01).
5) Run the Survival Analysis and Log Rank** blocks.
6) Enter genotypes of interest and run next block to plot them on with Kaplan–Meier curve.
7) Save all outputs to external files.

*Censoring should be used for escaped, lost, or otherwise unobserved flies, not necessarily for all missing data. Missing observations should be investigated rather than automatically treated as censored.

**Please note that the log-rank test does not inherently account for multiple comparisons. When multiple genotypes are compared simultaneously, consider multiple-testing corrections to account for the increased probability of false-positive results. However, as a rule, lifespan experiments should be independently replicated, as variation in experimental conditions can substantially influence lifespan outcomes. Replication of significant results across independent experiments, using a stringent significance threshold ($P < 0.01$), helps establish the robustness of observed effects and reduce the likelihood of false-positive findings.

Happy fly flipping! 😊